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# Logic Design

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## Ch 4

# Karnaugh Maps

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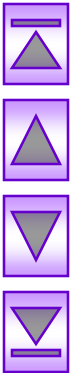
**IT Department**

**Al Balqa Applied University**



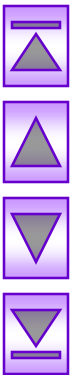
# Karnaugh Maps

- Introduction
- Venn Diagrams
- 2-variable K-maps
- 3-variable K-maps
- 4-variable K-maps
- 5-variable and larger K-maps
- Simplification using K-maps



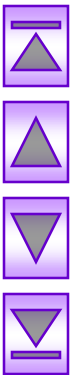
# Karnaugh Maps

- Converting to Minterms Form
- Simplest SOP Expressions
- Getting POS Expressions
- Don't-care Conditions
- Review
- Examples



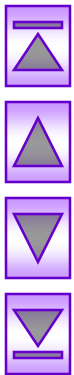
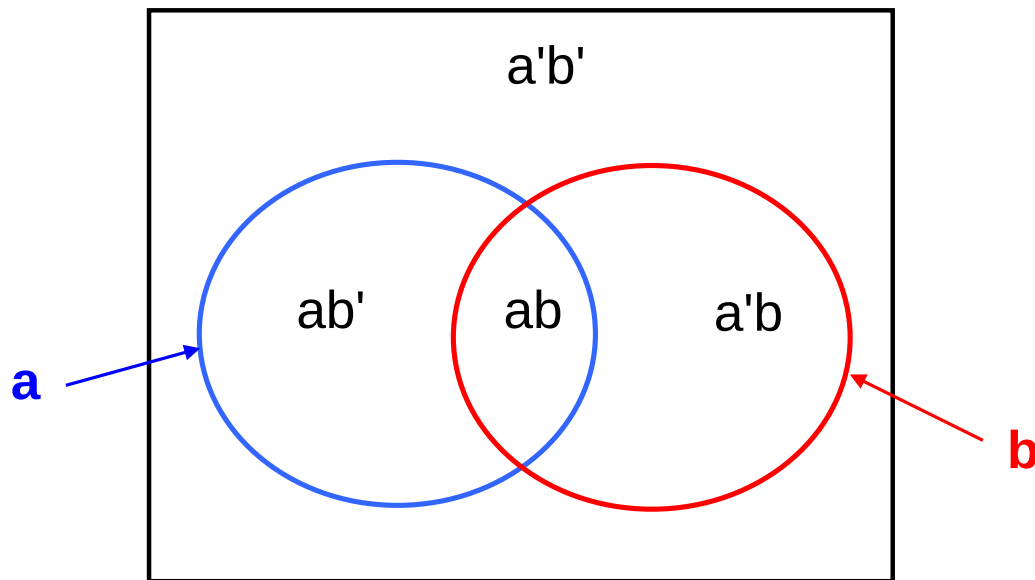
# Introduction

- Systematic method to obtain *simplified sum-of-products* (SOPs) Boolean expressions.
- Objective: *Fewest* possible terms/literals.
- Diagrammatic technique based on a special form of *Venn diagram*.
- Advantage: Easy with visual aid.
- Disadvantage: Limited to 5 or 6 variables.



# Venn Diagrams

- Venn diagram to represent the space of minterms.
- Example of 2 variables (4 minterms):



# Venn Diagrams

- Each set of minterms represents a Boolean function.  
Examples:

$$\{ ab, ab' \} \rightarrow ab + ab' = a(b+b') = a$$

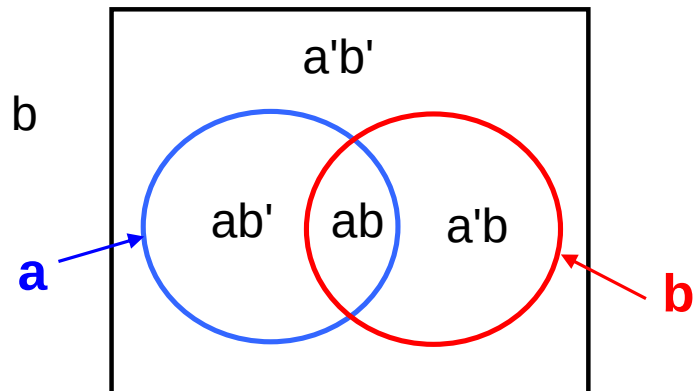
$$\{ a'b, ab \} \rightarrow a'b + ab = (a'+a)b = b$$

$$\{ ab \} \rightarrow ab$$

$$\{ ab, ab', a'b \} \rightarrow ab + ab' + a'b = a + b$$

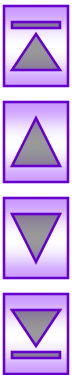
$$\{ \} \rightarrow 0$$

$$\{ a'b', ab, ab', a'b \} \rightarrow 1$$



# 2-variable K-maps

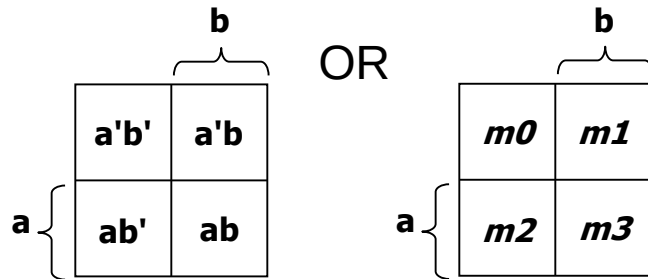
- **Karnaugh-map** (K-map) is an abstract form of Venn diagram, organised as a matrix of squares, where
  - ❖ each square represents a **minterm**
  - ❖ adjacent squares always **differ by just one literal** (so that the unifying theorem may apply:  $a + a' = 1$ )
- For 2-variable case (e.g.: variables a,b), the map can be drawn as:



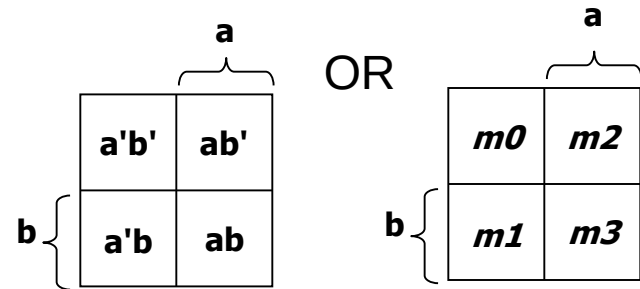
# 2-variable K-maps

- Alternative layouts of a 2-variable (a, b) K-map

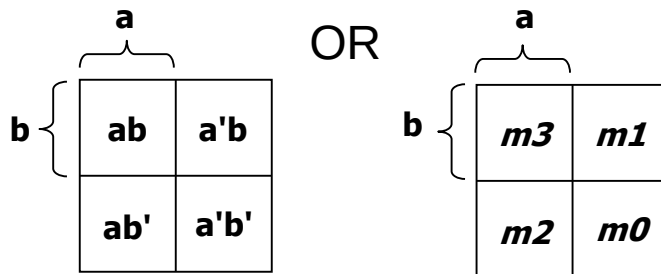
Alternative 1:



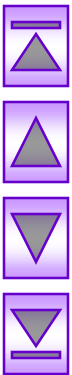
Alternative 2:



Alternative 3:



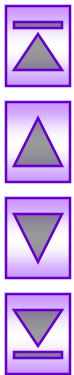
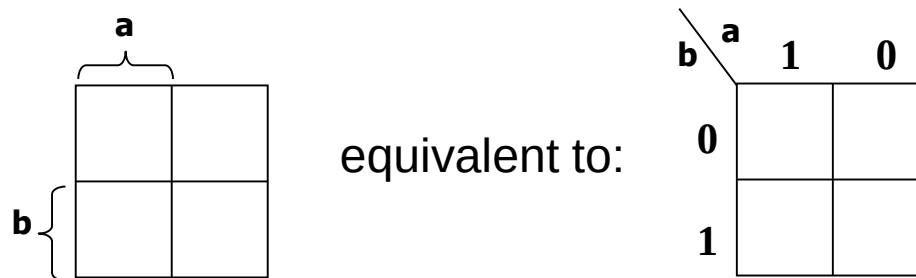
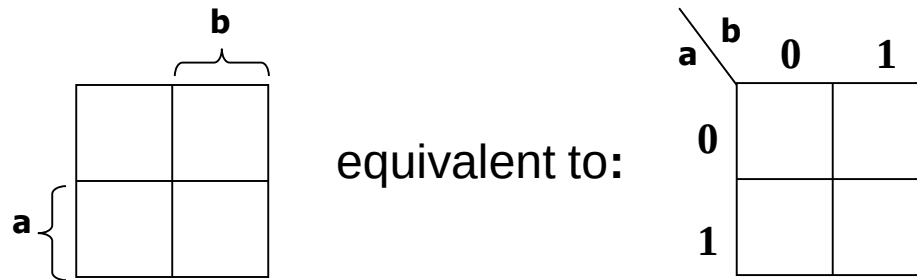
and others...





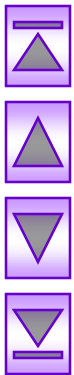
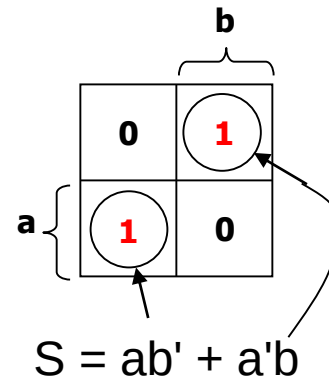
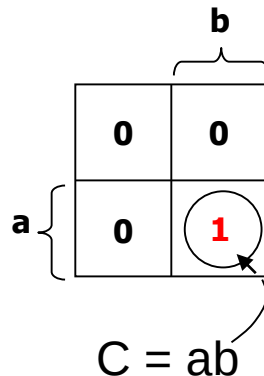
# 2-variable K-maps

- Equivalent labeling:



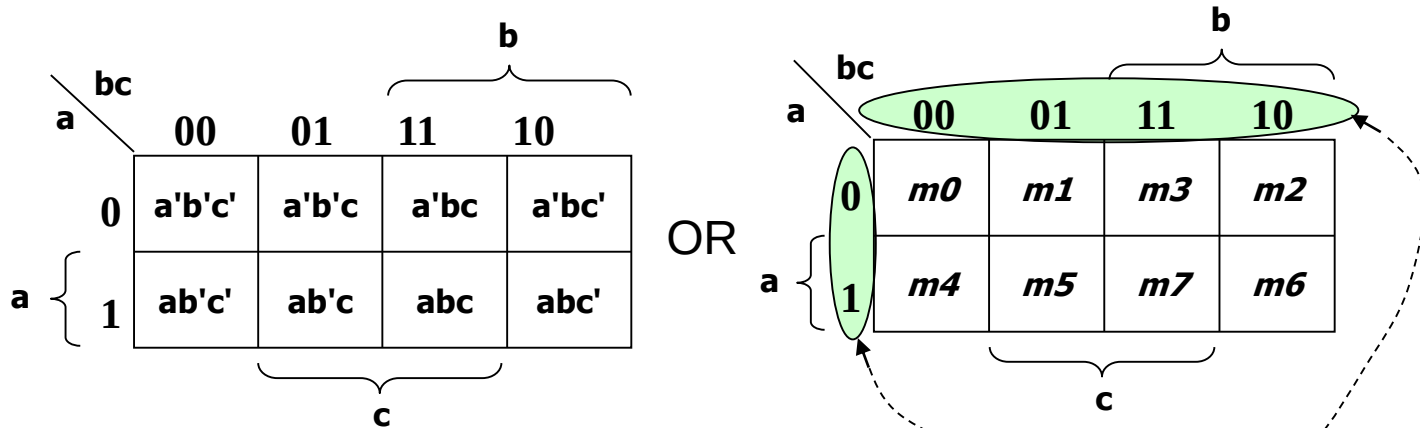
# 2-variable K-maps

- The K-map for a function is specified by putting
  - ❖ a '1' in the square corresponding to a minterm
  - ❖ a '0' otherwise
- For example: Carry and Sum of a half adder.



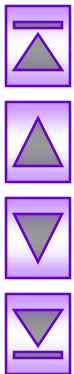
# 3-variable K-maps

- There are 8 minterms for 3 variables (a, b, c). Therefore, there are 8 cells in a 3-variable K-map.



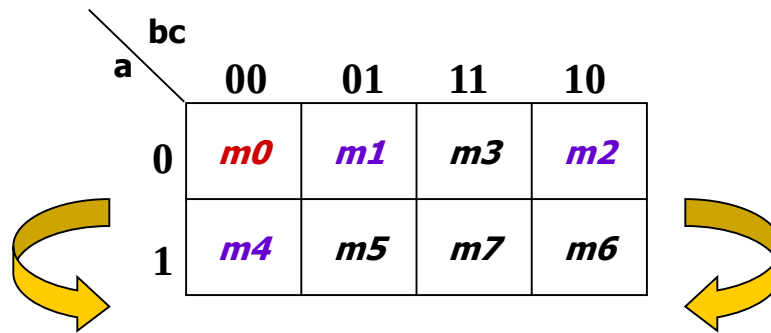
Above arrangement ensures that minterms of adjacent cells **differ by only ONE literal**. (Other arrangements which satisfy this criterion may also be used.)

**Note Gray code sequence**

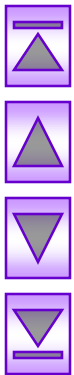


# 3-variable K-maps

- There is **wrap-around** in the K-map:
  - ❖  $a'b'c'$  ( $m0$ ) is adjacent to  $a'bc'$  ( $m2$ )
  - ❖  $ab'c'$  ( $m4$ ) is adjacent to  $abc'$  ( $m6$ )



Each cell in a 3-variable K-map has 3 adjacent neighbours. In general, each cell in an  $n$ -variable K-map has  $n$  adjacent neighbours. For example,  $m0$  has 3 adjacent neighbours:  $m1$ ,  $m2$  and  $m4$ .



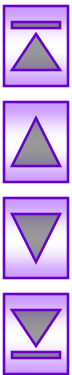
# Solve it yourself (Exercise 6.1)

1. The K-map of a 3-variable function  $F$  is shown below. What is the sum-of-minterms expression of  $F$ ?

|   |    | b  |    |    |    |
|---|----|----|----|----|----|
|   |    | c  |    |    |    |
| a | bc | 00 | 01 | 11 | 10 |
|   | 0  | 1  | 0  | 0  | 1  |
| 1 | 1  | 0  | 1  | 0  | 0  |

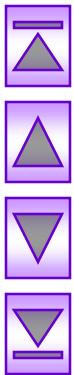
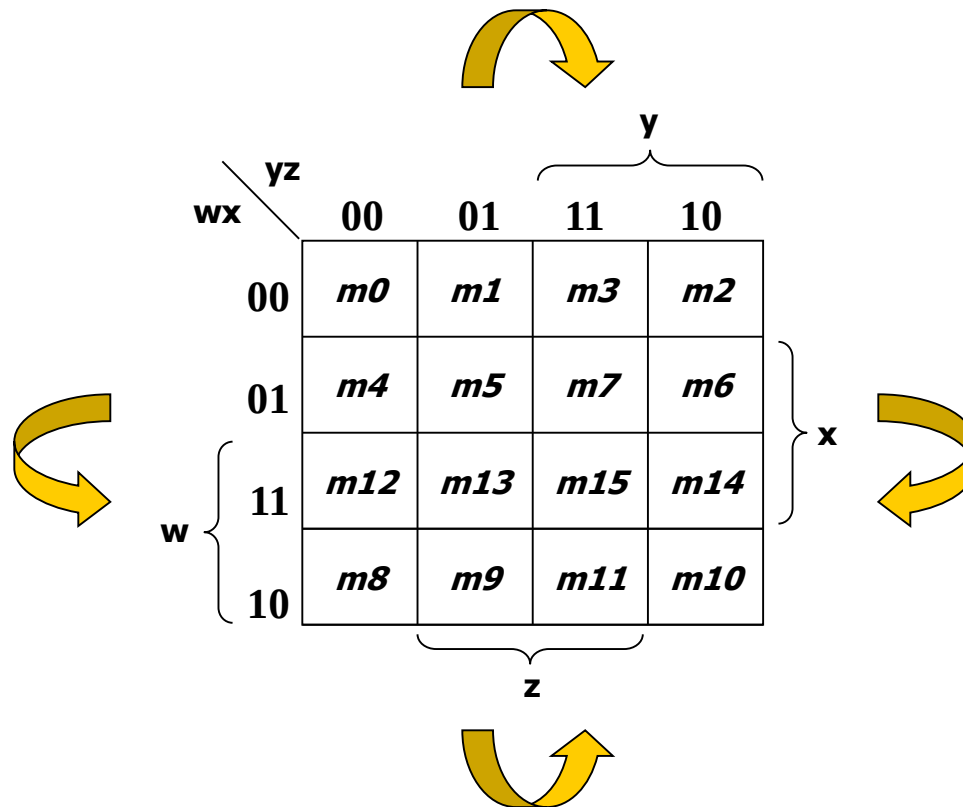
2. Draw the K-map for this function A:

$$A(x, y, z) = x.y + y.z' + x'.y'.z$$



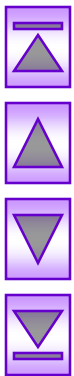
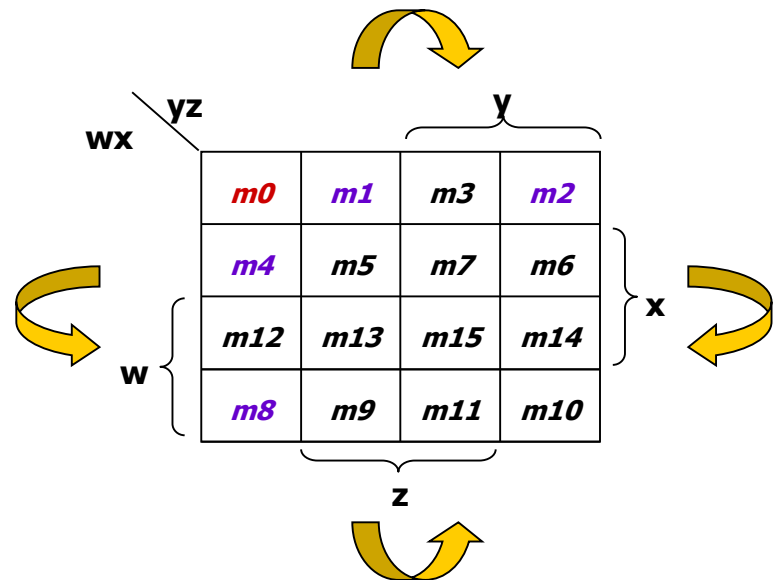
# 4-variable K-maps

- There are 16 cells in a 4-variable (w, x, y, z) K-map.



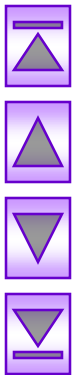
# 4-variable K-maps

- There are 2 wrap-arounds: a horizontal wrap-around and a vertical wrap-around.
- Every cell thus has 4 neighbours. For example, the cell corresponding to minterm  $m0$  has neighbours  $m1$ ,  $m2$ ,  $m4$  and  $m8$ .



# 5-variable K-maps

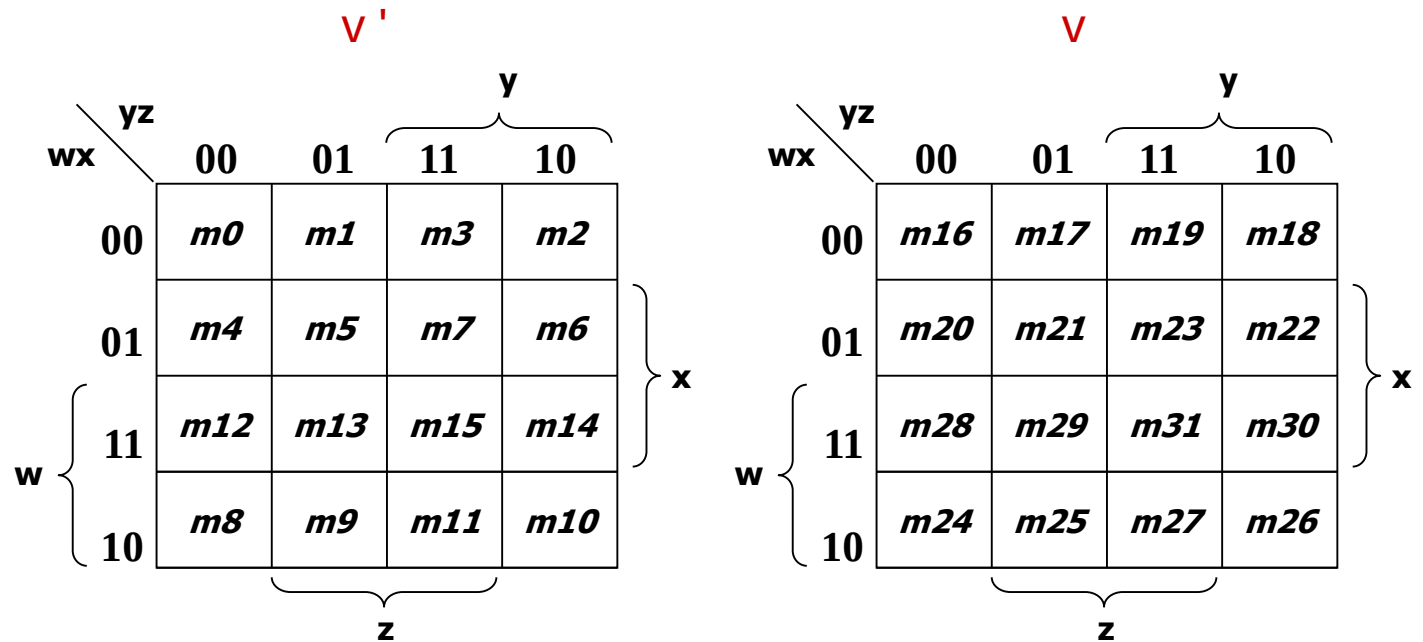
- Maps of more than 4 variables are more difficult to use because the geometry (hyper-cube configurations) for combining adjacent squares becomes more involved.
- For 5 variables, e.g. vwxyz, need  $2^5 = 32$  squares.



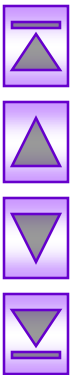


# 5-variable K-maps

- Organised as two 4-variable K-maps:

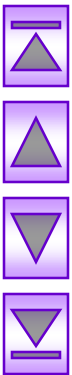


Corresponding squares of each map are adjacent.  
 Can visualise this as being *one 4-variable map on TOP of the other 4-variable map.*

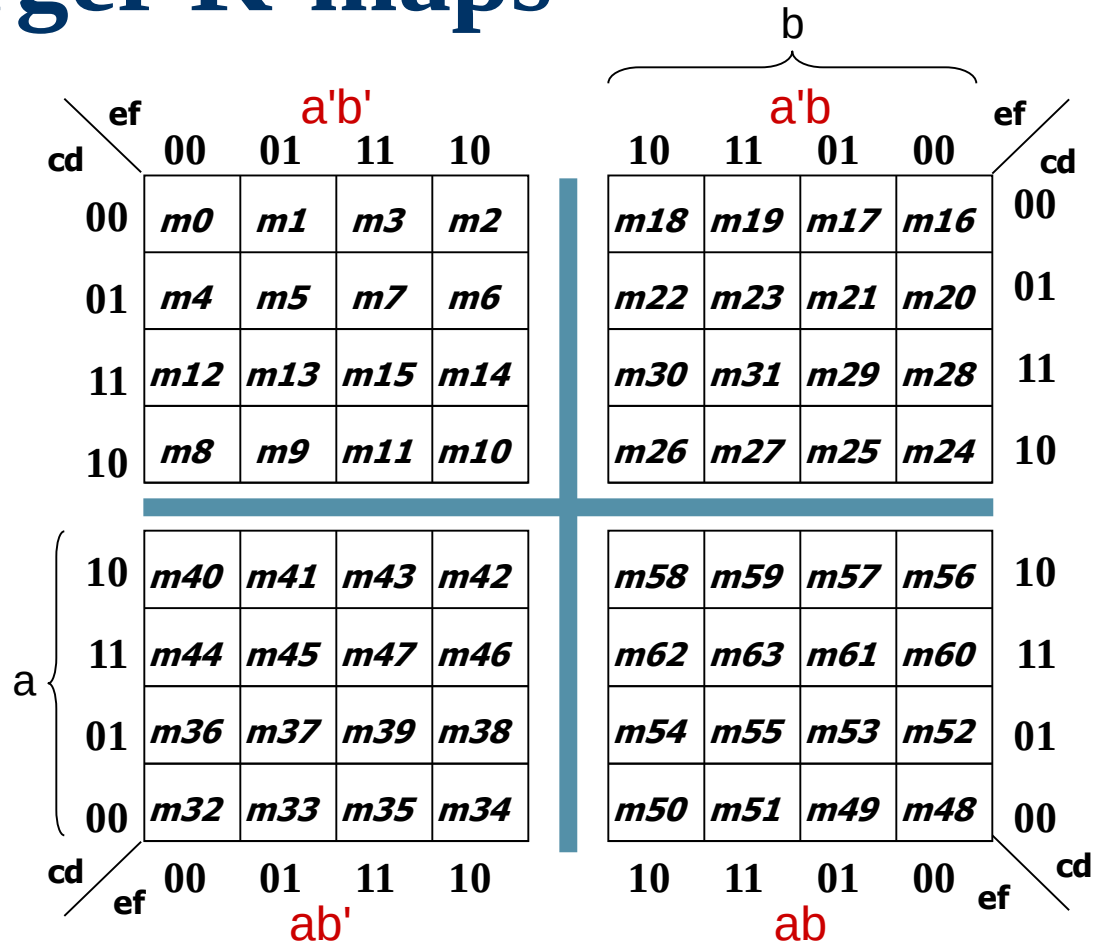


# Larger K-maps

- 6-variable K-map is pushing the limit of human “pattern-recognition” capability.
- K-maps larger than 6 variables are practically unheard of!
- Normally, a 6-variable K-map is organised as four 4-variable K-maps, which are mirrored along two axes.



# Larger K-maps



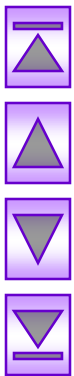
Try stretch your recognition capability by finding simplest sum-of-products expression for  $\Sigma m(6,8,14,18,23,25,27,29,41,45,57,61)$ .

# Simplification Using K-maps

- Based on the **Unifying Theorem**:

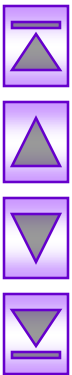
$$A + A' = 1$$

- In a K-map, each cell containing a '1' corresponds to a minterm of a given function  $F$ .
- Each group of adjacent cells containing '1' (group must have size *in powers of twos*: 1, 2, 4, 8, ...) then corresponds to a **simpler product term** of  $F$ .
  - ❖ Grouping 2 adjacent squares eliminates 1 variable, grouping 4 squares eliminates 2 variables, grouping 8 squares eliminates 3 variables, and so on. In general, grouping  $2^n$  squares eliminates  $n$  variables.



# Simplification Using K-maps

- Group as many squares as possible.
  - ❖ The larger the group is, the fewer the number of literals in the resulting product term.
- Select as few groups as possible to cover all the squares (minterms) of the function.
  - ❖ The fewer the groups, the fewer the number of product terms in the minimized function.



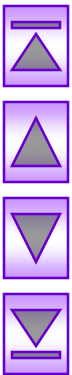
# Simplification Using K-maps

- Example:

$$\begin{aligned} F(w,x,y,z) &= w'xy'z' + w'xy'z + wx'yz' \\ &\quad + wx'yz + wxyz' + wxyz \\ &= \Sigma m(4, 5, 10, 11, 14, 15) \end{aligned}$$

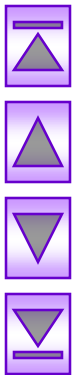
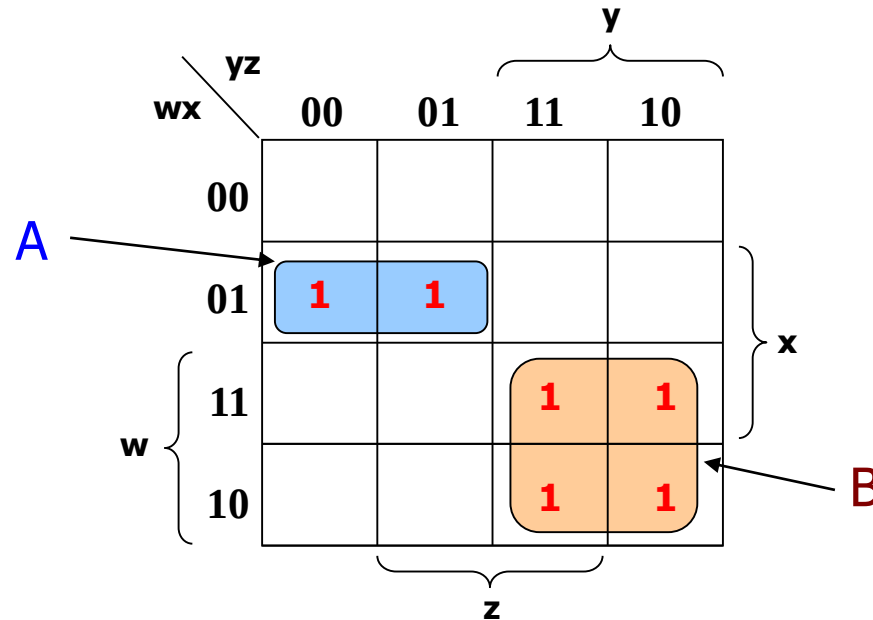
|   |    | y  |    |    |    |   |
|---|----|----|----|----|----|---|
|   |    | 00 | 01 | 11 | 10 |   |
| w | yz | 00 | 01 | 11 | 10 | x |
|   | 00 |    |    |    |    |   |
|   | 01 | 1  | 1  |    |    |   |
|   | 11 |    |    | 1  | 1  |   |
|   | 10 |    |    | 1  | 1  | z |

(cells with '0' are not shown for clarity)



# Simplification Using K-maps

- Each group of adjacent minterms (group size in powers of twos) corresponds to a possible **product term** of the given function.

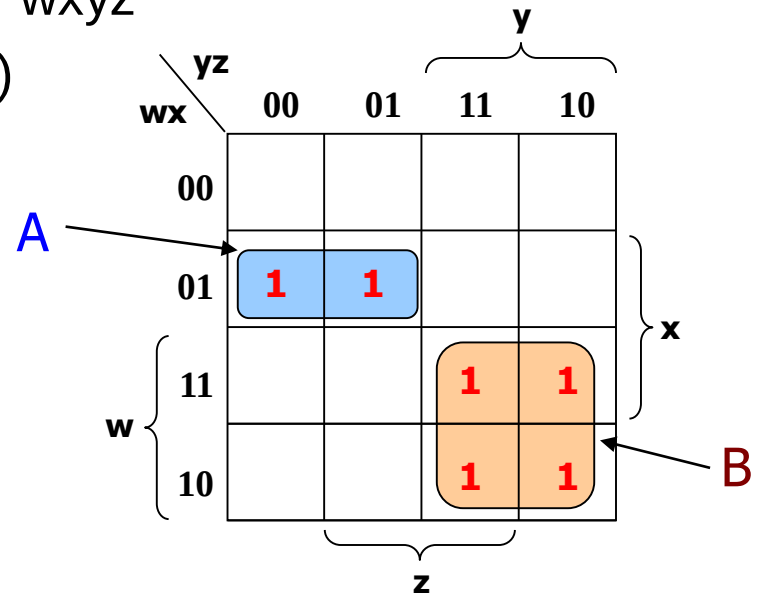


# Simplification Using K-maps

- There are 2 groups of minterms: A and B, where:

$$\begin{aligned}
 A &= w'xy'z' + w'xy'z \\
 &= w'xy'(z' + z) \\
 &= w'xy'
 \end{aligned}$$

$$\begin{aligned}
 B &= wx'yz' + wx'yz + wxyz' + wxyz \\
 &= wx'y(z' + z) + wxy(z' + z) \\
 &= wx'y + wxy \\
 &= w(x' + x)y \\
 &= wy
 \end{aligned}$$

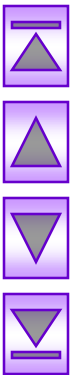




# Simplification Using K-maps

- Each product term of a group,  $w'xy'$  and  $wy$ , represents the *sum of minterms* in that group.
- Boolean function is therefore the sum of product terms (SOP) which represent all groups of the minterms of the function.

$$F(w,x,y,z) = A + B = w'xy' + wy$$



# Simplification Using K-maps

- Larger groups correspond to product terms of fewer literals. In the case of a 4-variable K-map:

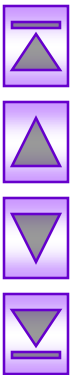
1 cell = 4 literals, e.g.:  $wxyz$ ,  $w'xy'z$

2 cells = 3 literals, e.g.:  $wxy$ ,  $wy'z'$

4 cells = 2 literals, e.g.:  $wx$ ,  $x'y$

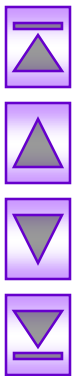
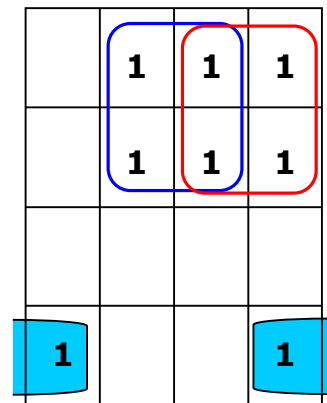
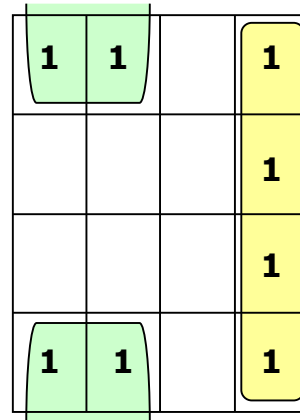
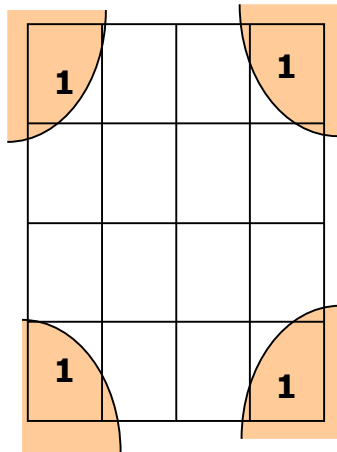
8 cells = 1 literal, e.g.:  $w$ ,  $y'$ ,  $z$

16 cells = no literal, e.g.:  $1$



# Simplification Using K-maps

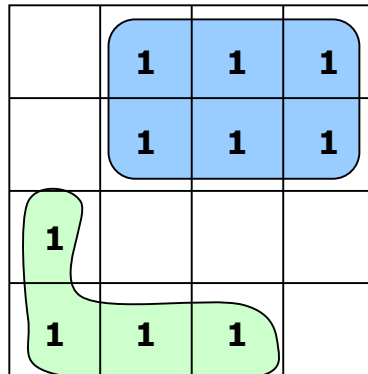
- Other possible valid groupings of a 4-variable K-map include:



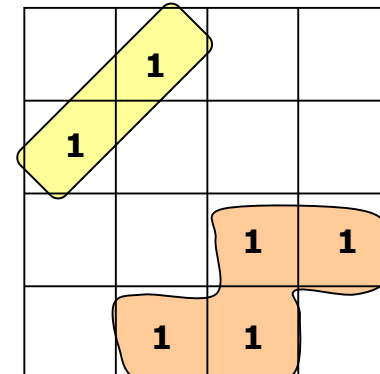
# Simplification Using K-maps

- Groups of minterms must be
  - (1) rectangular, and
  - (2) have size in powers of 2's.

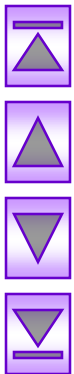
Otherwise they are *invalid* groups. Some examples of *invalid groups*:



x

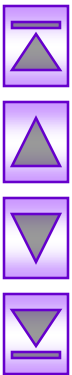


x



# Converting to Minterms Form

- The K-map of a function is easily drawn when the function is given in canonical sum-of-products, or sum-of-minterms form.
- What if the function is not in sum-of-minterms?
  - ❖ Convert it to sum-of-products (SOP) form.
  - ❖ Expand the SOP expression into sum-of-minterms expression, or fill in the K-map directly based on the SOP expression.

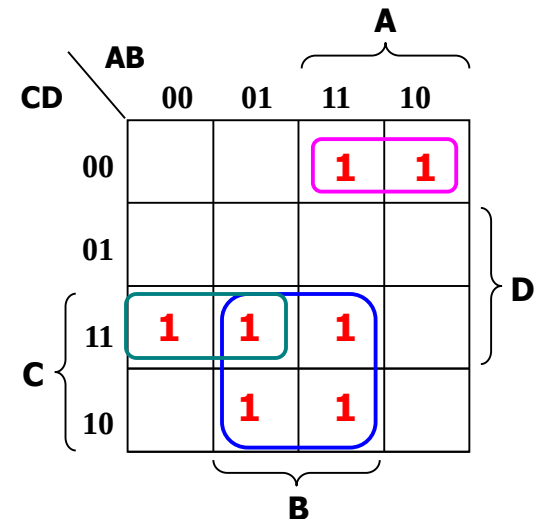


# Converting to Minterms Form

## ■ Example:

$$\begin{aligned}
 f(A,B,C,D) &= A(C+D)'(B'+D') + C(B+C'+A'D) \\
 &= A(C'D')(B'+D') + BC + CC' + A'CD \\
 &= \underline{AB'C'D'} + \underline{AC'D'} + \underline{BC} + \underline{A'CD}
 \end{aligned}$$

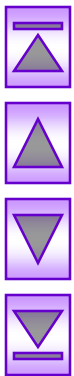
$$\begin{aligned}
 &AB'C'D' + \textcolor{red}{AC'D'} + BC + A'CD \\
 &= AB'C'D' + AC'D'(B+B') + \textcolor{red}{BC} + A'CD \\
 &= \underline{AB'C'D'} + \underline{ABC'D'} + \underline{AB'C'D'} + \\
 &\quad BC(A+A') + A'CD \\
 &= AB'C'D' + ABC'D' + ABC + A'BC + \\
 &\quad A'CD \\
 &= AB'C'D' + ABC'D' + ABC(D+D') + \\
 &\quad A'BC(D+D') + A'CD(B+B') \\
 &= AB'C'D' + ABC'D' + ABCD + ABCD' + \\
 &\quad A'BCD + A'BCD' + A'B'CD
 \end{aligned}$$



# Simplest SOP Expressions

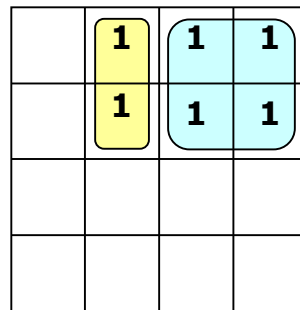
- To find the simplest possible *sum of products* (SOP) expression from a K-map, you need to obtain:
  - ❖ minimum number of literals per product term; and
  - ❖ minimum number of product terms
- This is achieved in K-map using
  - ❖ *bigger groupings* of minterms (**prime implicants**) where possible; and
  - ❖ *no redundant groupings* (look for **essential prime implicants**)

**Implicant:** a product term that could be used to cover minterms of the function.

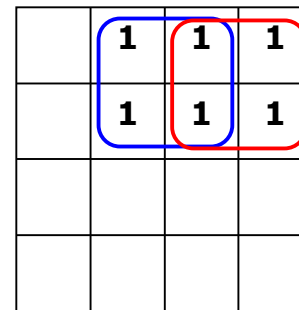


# Simplest SOP Expressions

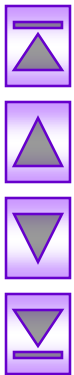
- A *prime implicant* is a product term obtained by combining the maximum possible number of minterms from adjacent squares in the map.
- Use bigger groupings (prime implicants) where possible.



x



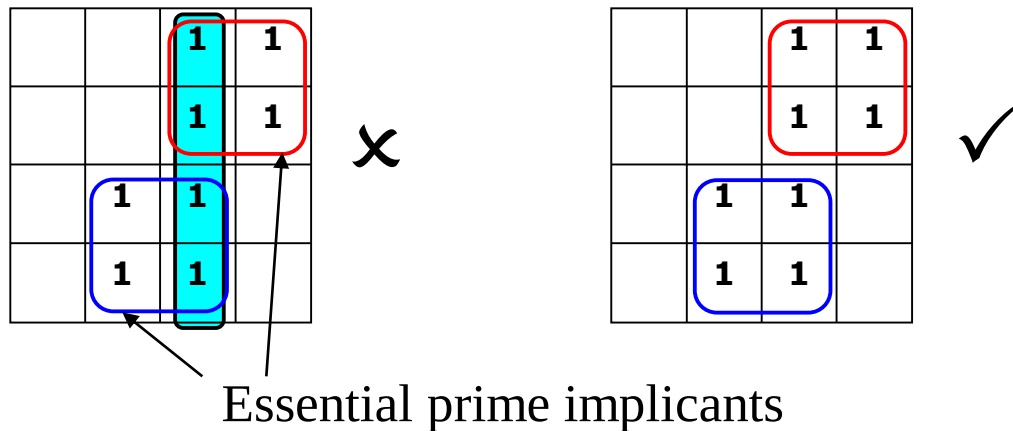
✓



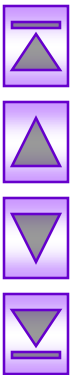


# Simplest SOP Expressions

- No redundant groups:



- An *essential prime implicant* is a prime implicant that includes at least one minterm that is not covered by any other prime implicant.

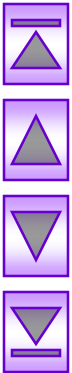


# Solve it yourself (Exercise 6.2)

Q. Identify the prime implicants and the essential prime implicants of the two K-maps below.

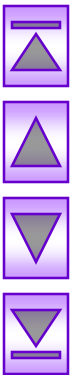
|   |    | b  |    |    |    |
|---|----|----|----|----|----|
|   |    | 00 | 01 | 11 | 10 |
| a | bc |    |    |    |    |
|   |    | 00 | 01 | 11 | 10 |
| 0 | 1  | 1  | 0  | 1  |    |
| 1 | 0  | 1  | 0  | 0  |    |

|    |    | A  |    |    |    |
|----|----|----|----|----|----|
|    |    | 00 | 01 | 11 | 10 |
| CD | AB |    |    |    |    |
|    |    | 00 | 01 | 11 | 10 |
| 00 | 1  | 1  |    | 1  |    |
| 01 |    |    | 1  | 1  |    |
| 11 | 1  | 1  |    | 1  |    |
| 10 | 1  |    | 1  | 1  |    |



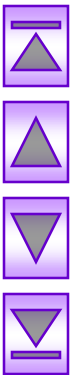
# Simplest SOP Expressions

- Algorithm 1 (non optimal):
  1. Count the number of adjacencies for each minterm on the K-map.
  2. Select an uncovered minterm with the fewest number of adjacencies. Make an arbitrary choice if more than one choice is possible.
  3. Generate a prime implicant for this minterm and put it in the cover. If this minterm is covered by more than one prime implicant, select the one that covers the most uncovered minterms.
  4. Repeat steps 2 and 3 until all the minterms have been covered.



# Simplest SOP Expressions

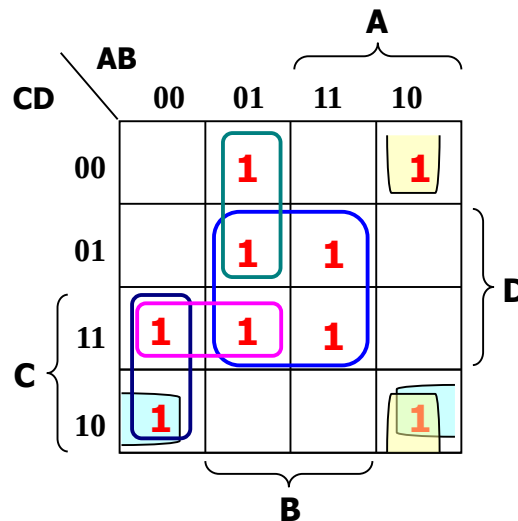
- Algorithm 2 (non optimal):
  1. Circle all prime implicants on the K-map.
  2. Identify and select all essential prime implicants for the cover.
  3. Select a minimum subset of the remaining prime implicants to complete the cover, that is, to cover those minterms not covered by the essential prime implicants.



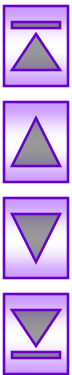
# Simplest SOP Expressions

- Example:

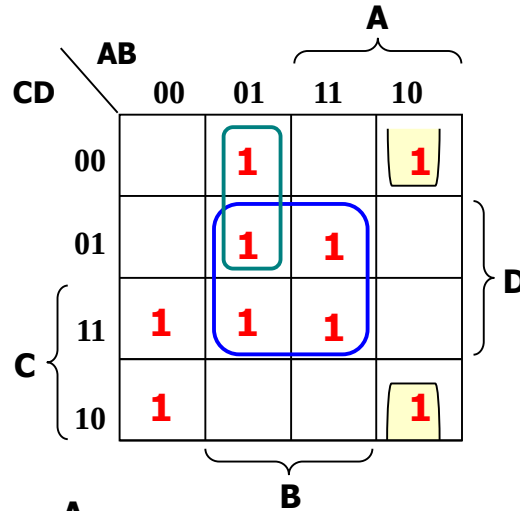
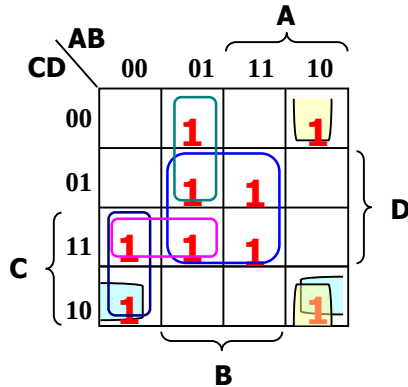
$$f(A,B,C,D) = \sum m(2,3,4,5,7,8,10,13,15)$$



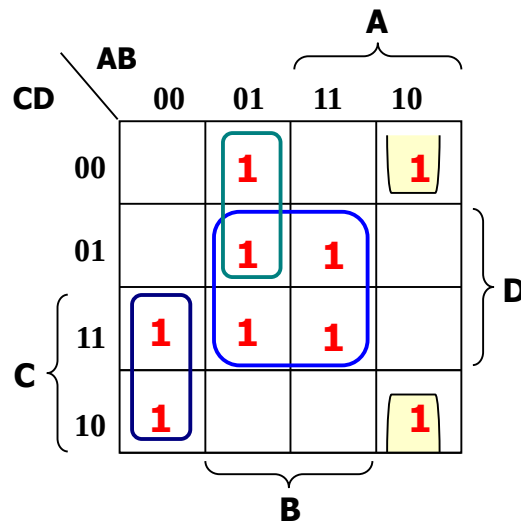
All prime implicants



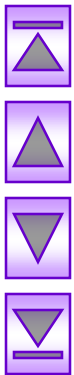
# Simplest SOP Expressions



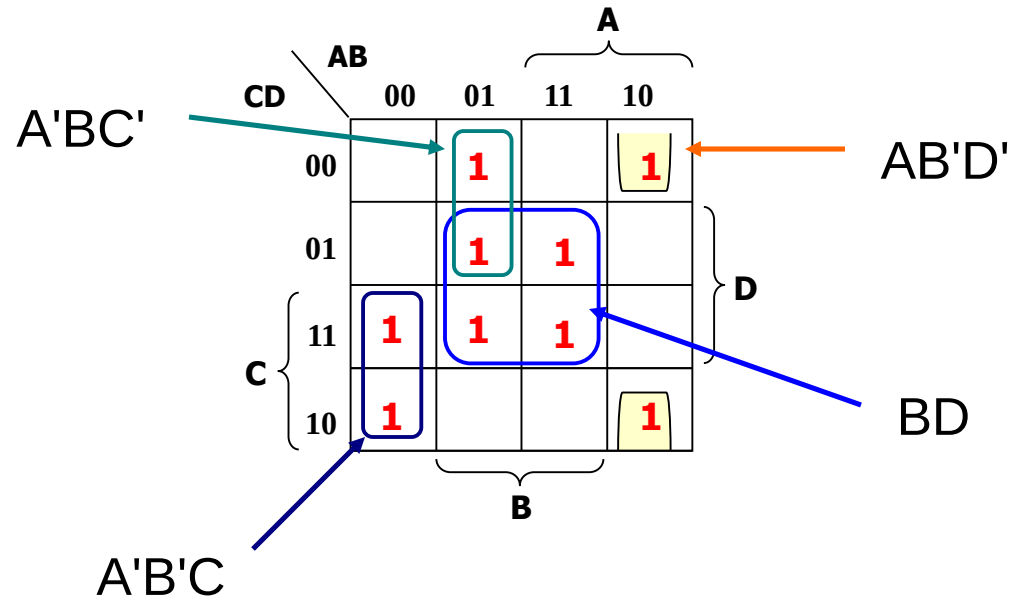
← Essential prime implicants



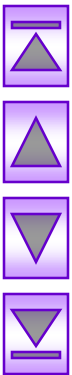
← Minimum cover



# Simplest SOP Expressions



$$f(A,B,C,D) = BD + A'B'C + AB'D' + A'BC'$$



# Solve it yourself (Exercise 6.3)

Q. Find the simplified expression for  $G(A,B,C,D)$ .

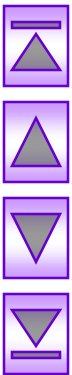
|    |    | A  |   |    |    |
|----|----|----|---|----|----|
|    |    | AB |   | 11 | 10 |
| CD | 00 |    | 1 |    |    |
|    | 01 |    | 1 | 1  | 1  |
| C  | 11 | 1  | 1 | 1  |    |
|    | 10 |    |   | 1  |    |

Diagram illustrating a 4x4 Karnaugh map for the function  $G(A,B,C,D)$ . The map is labeled with variables A, B, C, and D. The columns are labeled AB (00, 01, 11, 10) and the rows are labeled CD (00, 01, 11, 10). The map shows the following cells containing 1s:

- Row 00: Column 01 (1)
- Row 01: Columns 01 (1), 11 (1), 10 (1)
- Row 11: Columns 00 (1), 01 (1), 11 (1)
- Row 10: Column 11 (1)

Groupings are indicated by brackets:

- Group A: Columns 11 and 10 (covering cells (01, 11), (01, 10), (11, 11), (11, 10)).
- Group B: Columns 01 and 11 (covering cells (00, 01), (00, 11), (01, 01), (01, 11), (11, 01), (11, 11), (10, 01), (10, 11)).
- Group C: Rows 11 and 10 (covering cells (11, 00), (11, 01), (11, 11), (11, 10), (10, 00), (10, 01), (10, 11), (10, 10)).
- Group D: Rows 01 and 11 (covering cells (01, 00), (01, 01), (11, 00), (11, 01), (10, 00), (10, 01), (10, 11), (10, 10)).



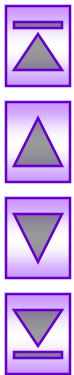


# Getting POS Expressions

- Simplified POS expression can be obtained by grouping the maxterms (i.e. 0s) of given function.
- Example:

Given  $F = \sum m(0,1,2,3,5,7,8,9,10,11)$ , we first draw the K-map, then group the maxterms together:

|    |    | A  |    |    |   |
|----|----|----|----|----|---|
|    |    | AB |    |    |   |
| CD | 00 | 01 | 11 | 10 |   |
|    | 00 | 1  | 0  | 0  | 1 |
| 01 | 1  | 1  | 0  | 1  | D |
| 11 | 1  | 1  | 0  | 1  |   |
| 10 | 1  | 0  | 0  | 1  |   |
| 00 | 1  | 0  | 0  | 1  |   |
|    |    | B  |    |    |   |



# Getting POS Expressions

K-map of  $F$

|    |    | A  |    |    |    |
|----|----|----|----|----|----|
|    |    | AB |    |    |    |
| CD |    | 00 | 01 | 11 | 10 |
|    | 00 | 1  | 0  | 0  | 1  |
| C  | 01 | 1  | 1  | 0  | 1  |
|    | 11 | 1  | 1  | 0  | 1  |
|    | 10 | 1  | 0  | 0  | 1  |
|    |    |    |    |    |    |
|    |    | B  |    | D  |    |

K-map of  $F'$

|    |    | A  |    |    |    |
|----|----|----|----|----|----|
|    |    | AB |    |    |    |
| CD |    | 00 | 01 | 11 | 10 |
|    | 00 | 0  | 1  | 1  | 0  |
| C  | 01 | 0  | 0  | 1  | 0  |
|    | 11 | 0  | 0  | 1  | 0  |
|    | 10 | 0  | 1  | 1  | 0  |
|    |    |    |    |    |    |
|    |    | B  |    | D  |    |

- This gives the SOP of  $F'$  to be:

$$F' = BD' + AB$$

- To get POS of  $F$ , we have:

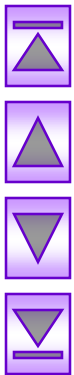
$$F = (BD' + AB)'$$

$$= (BD')'(AB)'$$

DeMorgan

$$= (B' + D)(A' + B')$$

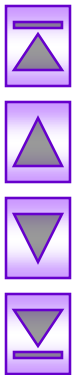
DeMorgan



# Don't-care Conditions

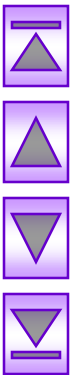
- In certain problems, some outputs are not specified.
- These outputs can be either '1' or '0'.
- They are called **don't-care conditions**, denoted by X (or sometimes, d).
- Example: An odd parity generator for BCD code which has 6 unused combinations.

| No. | A | B | C | D | P |
|-----|---|---|---|---|---|
| 0   | 0 | 0 | 0 | 0 | 1 |
| 1   | 0 | 0 | 0 | 1 | 0 |
| 2   | 0 | 0 | 1 | 0 | 0 |
| 3   | 0 | 0 | 1 | 1 | 1 |
| 4   | 0 | 1 | 0 | 0 | 0 |
| 5   | 0 | 1 | 0 | 1 | 1 |
| 6   | 0 | 1 | 1 | 0 | 1 |
| 7   | 0 | 1 | 1 | 1 | 0 |
| 8   | 1 | 0 | 0 | 0 | 0 |
| 9   | 1 | 0 | 0 | 1 | 1 |
| 10  | 1 | 0 | 1 | 0 | X |
| 11  | 1 | 0 | 1 | 1 | X |
| 12  | 1 | 1 | 0 | 0 | X |
| 13  | 1 | 1 | 0 | 1 | X |
| 14  | 1 | 1 | 1 | 0 | X |
| 15  | 1 | 1 | 1 | 1 | X |



# Don't-care Conditions

- Don't-care conditions can be used to help simplify Boolean expression further in K-maps.
- They could be chosen to be either '1' or '0', depending on which gives the simpler expression.



# Don't-care Conditions

- For comparison:

❖ WITHOUT Don't-cares:

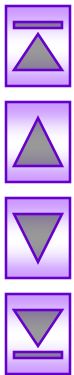
$$P = A'B'C'D' + A'B'CD + A'BC'D + A'BCD' + AB'C'D$$

❖ WITH Don't-cares:

$$P = A'B'C'D' + B'CD + BC'D + BCD' + AD$$

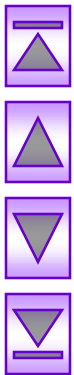
|    |    | C  |    |    |    |
|----|----|----|----|----|----|
|    |    | D  |    |    |    |
| AB | CD | 00 | 01 | 11 | 10 |
|    |    |    |    |    |    |
| A  | 00 | 1  |    | 1  |    |
|    | 01 |    | 1  |    | 1  |
|    | 11 |    |    |    |    |
|    | 10 |    | 1  |    |    |

|    |    | C  |    |    |    |
|----|----|----|----|----|----|
|    |    | D  |    |    |    |
| AB | CD | 00 | 01 | 11 | 10 |
|    |    |    |    |    |    |
| A  | 00 | 1  |    | 1  |    |
|    | 01 |    | 1  |    | 1  |
|    | 11 | X  | X  | X  | X  |
|    | 10 |    | 1  | X  | X  |



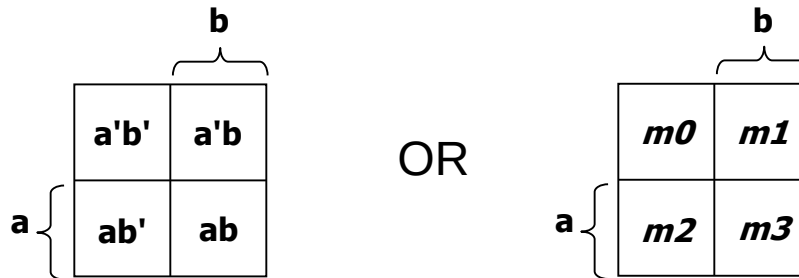
# Review – The Techniques

- Algebraic Simplification.
  - ❖ requires skill but extremely open-ended.
- Karnaugh Maps.
  - ❖ can obtain simplified standard forms.
  - ❖ easy for humans (pattern-matching skills).
  - ❖ limited to not more than 6 variables.
- Other computer-aided techniques such as Quine-McCluskey method (not covered in this course).



# Review – K-maps

- Characteristics of K-map layouts:
  - (i) each minterm in one square/cell
  - (ii) adjacent/neighbouring minterms differ by only 1 literal
  - (iii)  $n$ -literal minterm has  $n$  neighbours/adjacent cells
- Valid 2-, 3-, 4-variable K-maps



# Review – K-maps

|   |   | b        |         |        |         |
|---|---|----------|---------|--------|---------|
|   |   | bc       |         |        |         |
| a | 0 | $a'b'c'$ | $a'b'c$ | $a'bc$ | $a'bc'$ |
|   | 1 | $ab'c'$  | $ab'c$  | $abc$  | $abc'$  |

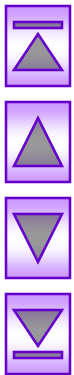
Diagram illustrating a 2x4 Karnaugh map for variables a, b, and c. The map is organized with rows labeled 'a' (0 and 1) and columns labeled 'bc' (00, 01, 11, 10). The cells contain the corresponding minterm expressions:  $a'b'c'$ ,  $a'b'c$ ,  $a'bc$ ,  $a'bc'$  for row 0, and  $ab'c'$ ,  $ab'c$ ,  $abc$ ,  $abc'$  for row 1. Brackets indicate the grouping of columns by 'b' and rows by 'a', and a bracket under the bottom row indicates grouping by 'c'.

|   |   | b    |      |      |      |
|---|---|------|------|------|------|
|   |   | bc   |      |      |      |
| a | 0 | $m0$ | $m1$ | $m3$ | $m2$ |
|   | 1 | $m4$ | $m5$ | $m7$ | $m6$ |

Diagram illustrating a 2x4 Karnaugh map for variables a, b, and c, showing minterm indices. The map is organized with rows labeled 'a' (0 and 1) and columns labeled 'bc' (00, 01, 11, 10). The cells contain the corresponding minterm indices:  $m0$ ,  $m1$ ,  $m3$ ,  $m2$  for row 0, and  $m4$ ,  $m5$ ,  $m7$ ,  $m6$  for row 1. Brackets indicate the grouping of columns by 'b' and rows by 'a', and a bracket under the bottom row indicates grouping by 'c'.

|    |    | y     |       |       |       |
|----|----|-------|-------|-------|-------|
|    |    | yz    |       |       |       |
| wx | 00 | $m0$  | $m1$  | $m3$  | $m2$  |
|    | 01 | $m4$  | $m5$  | $m7$  | $m6$  |
| w  | 11 | $m12$ | $m13$ | $m15$ | $m14$ |
|    | 10 | $m8$  | $m9$  | $m11$ | $m10$ |

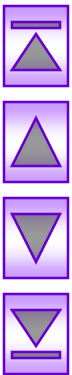
Diagram illustrating a 4x4 Karnaugh map for variables w, x, y, and z. The map is organized with rows labeled 'wx' (00, 01, 11, 10) and columns labeled 'yz' (00, 01, 11, 10). The cells contain the corresponding minterm indices:  $m0$ ,  $m1$ ,  $m3$ ,  $m2$  for row 00;  $m4$ ,  $m5$ ,  $m7$ ,  $m6$  for row 01;  $m12$ ,  $m13$ ,  $m15$ ,  $m14$  for row 11; and  $m8$ ,  $m9$ ,  $m11$ ,  $m10$  for row 10. Brackets indicate the grouping of columns by 'y' and rows by 'w', and a bracket under the bottom row indicates grouping by 'z'.





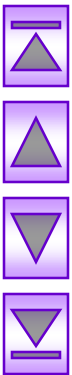
# Review – K-maps

- Groupings to select product-terms must be:
  - ❖ (i) rectangular in shape
  - ❖ (ii) in powers of twos (1, 2, 4, 8, etc.)
  - ❖ (iii) always select largest possible groupings of minterms (i.e. prime implicants)
  - ❖ (iv) eliminate redundant groupings
- Sum-of-products (SOP) form obtained by selecting groupings of minterms (corresponding to product terms).



# Review – K-maps

- Product-of-sums (POS) form obtained by selecting groupings of maxterms (corresponding to sum terms) and by applying DeMorgan's theorem.
- Don't cares, marked by X (or d), can denote either 1 or 0. They could therefore be selected as 1 or 0 to further simplify expressions.



# Examples

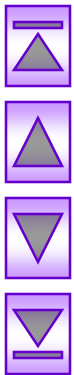
- Example #1:

$$f(A,B,C,D) = \sum m(2,3,4,5,7,8,10,13,15)$$

|    |    | A  |    |    |    |   |
|----|----|----|----|----|----|---|
|    |    | AB |    |    |    |   |
| CD |    | 00 | 01 | 11 | 10 |   |
|    | 00 |    | 1  |    | 1  |   |
|    | 01 |    | 1  | 1  |    | D |
| C  | 11 | 1  | 1  | 1  |    |   |
|    | 10 | 1  |    |    | 1  |   |
|    |    | B  |    |    |    |   |



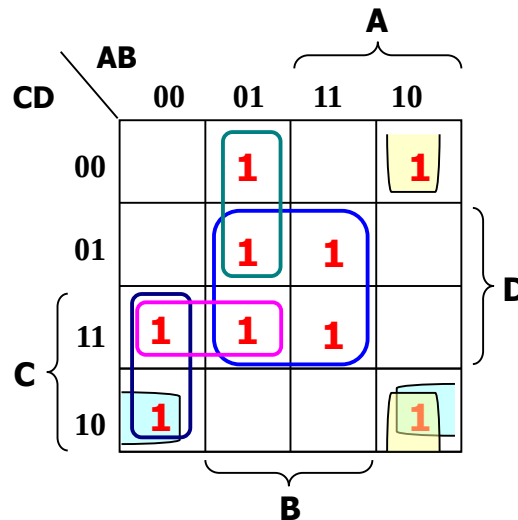
Fill in the 1's.



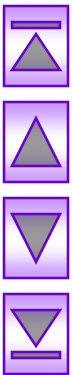
# Examples

- Example #1:

$$f(A,B,C,D) = \sum m(2,3,4,5,7,8,10,13,15)$$



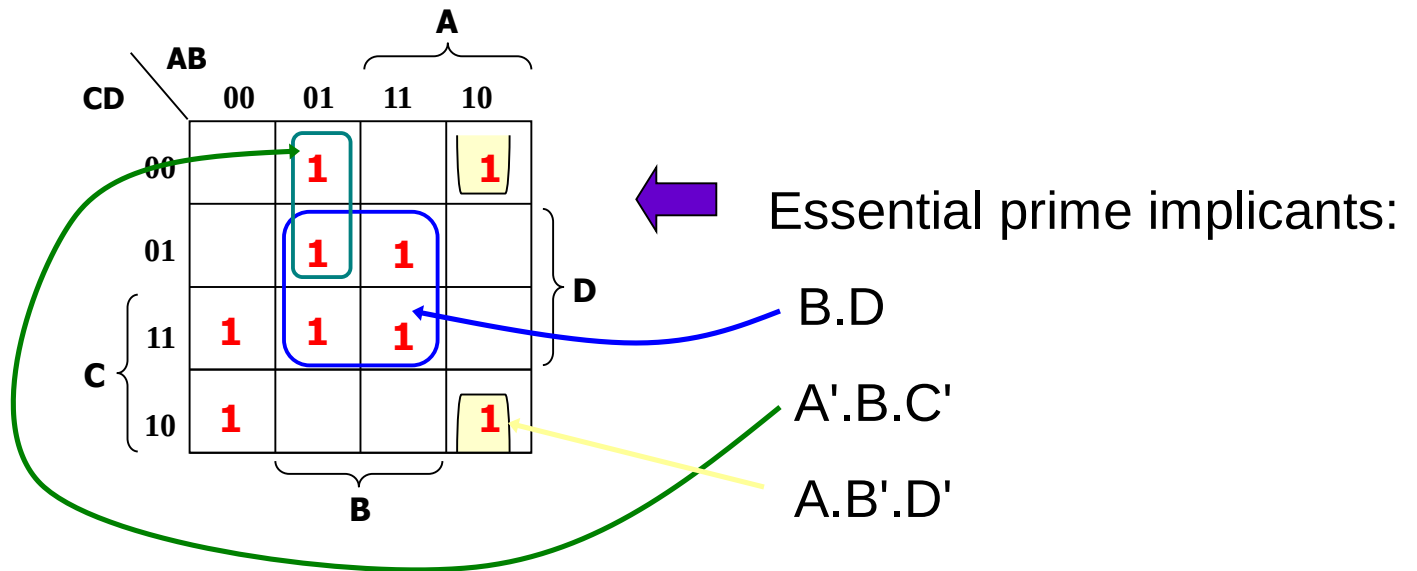
These are all the prime implicants; but do we need them all?



# Examples

- Example #1:

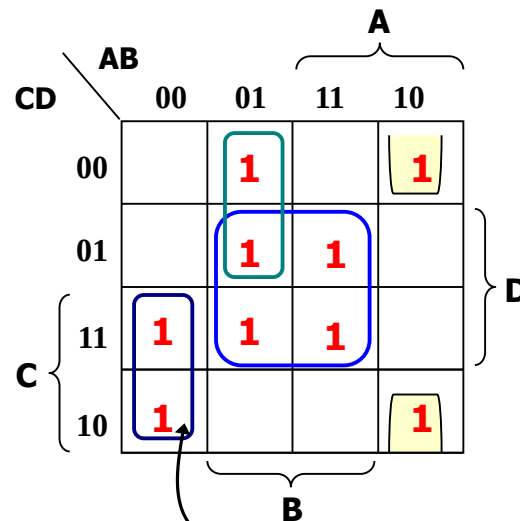
$$f(A,B,C,D) = \sum m(2,3,4,5,7,8,10,13,15)$$



# Examples

- Example #1:

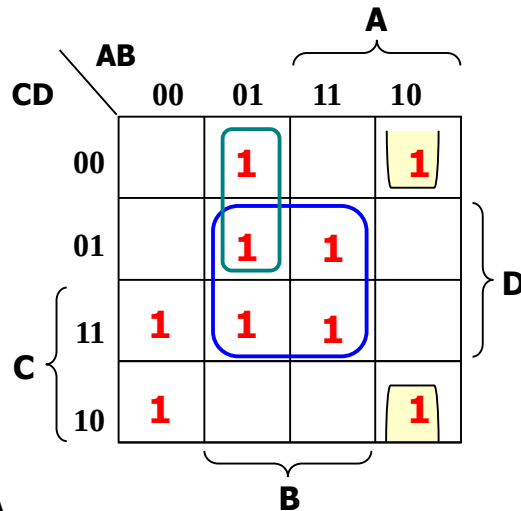
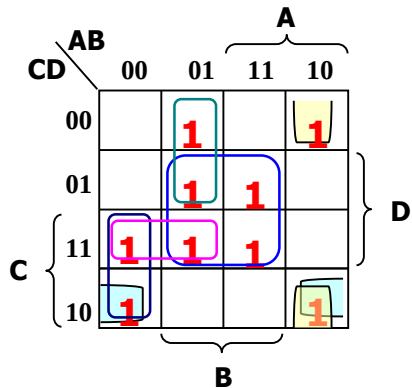
$$f(A,B,C,D) = \sum m(2,3,4,5,7,8,10,13,15)$$



← Minimum cover.  
 EPIs:  $B.D$ ,  $A'.B.C'$ ,  $A.B'.D'$   
 +  
 $A'.B'.C$

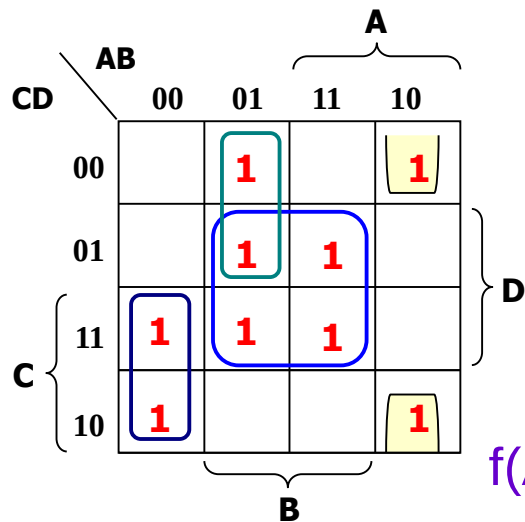
$$f(A,B,C,D) = B.D + A'.B.C' + A.B'.D' + A'.B'.C$$

# Examples



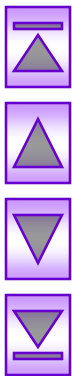
← Essential prime implicants

SUMMARY



← Minimum cover

$$f(A,B,C,D) = BD + A'B'C + AB'D' + A'B.C'$$



# Examples

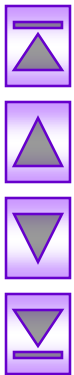
- Example #2:

$$f(A,B,C,D) = A.B.C + B'.C.D' + A.D + B'.C'.D'$$

|    |    | A  |    |    |    |
|----|----|----|----|----|----|
|    |    | AB |    |    |    |
| CD |    | 00 | 01 | 11 | 10 |
|    | 00 | 1  |    |    | 1  |
|    | 01 |    |    | 1  | 1  |
|    | 11 |    |    | 1  | 1  |
| C  | 10 | 1  |    | 1  | 1  |
|    |    | B  |    |    |    |



Fill in the 1's.

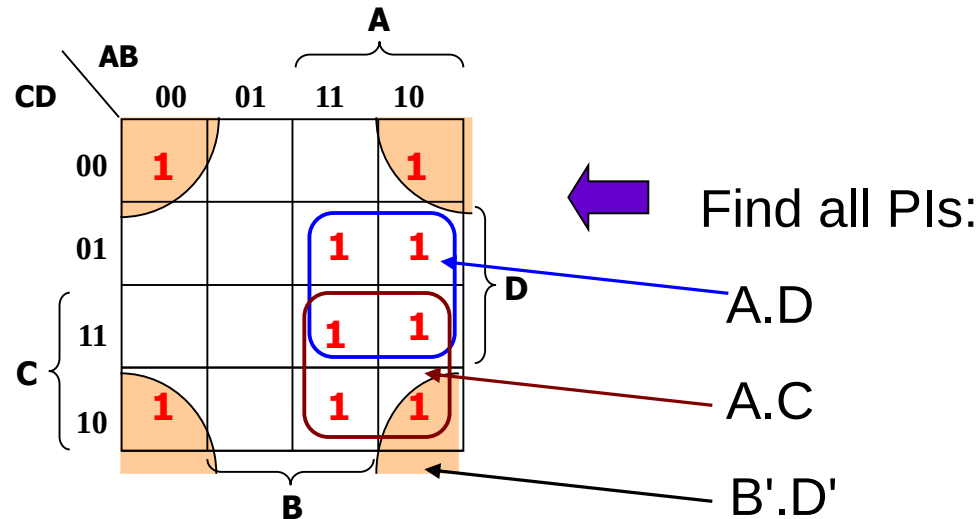




# Examples

- Example #2:

$$f(A,B,C,D) = A.B.C + B'.C.D' + A.D + B'.C'.D'$$



Are all '1's covered by the PIs? Yes, so the answer is:  $f(A,B,C,D) = A.D + A.C + B'.D'$

# Examples

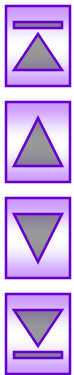
- Example #3 (with don't cares):

$$f(A,B,C,D) = \sum m(2,8,10,15) + \sum d(0,1,3,7)$$

| CD \ AB |    | A  |    |    |    |
|---------|----|----|----|----|----|
|         |    | 00 | 01 | 11 | 10 |
| C       | 00 | X  |    |    | 1  |
|         | 01 | X  |    |    |    |
|         | 11 | X  | X  | 1  |    |
|         | 10 | 1  |    |    | 1  |



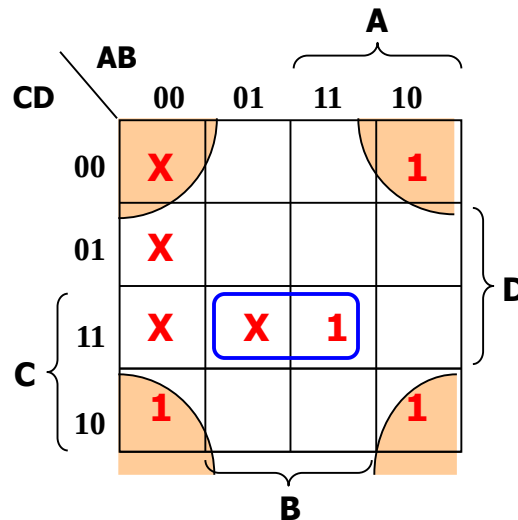
Fill in the 1's and X's.



# Examples

- Example #3 (with don't cares):

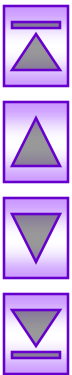
$$f(A,B,C,D) = \sum m(2,8,10,15) + \sum d(0,1,3,7)$$



Do we need to have an additional term  $A'.B'$  to cover the 2 remaining x's?

No, because all the 1's (minterms) have been covered.

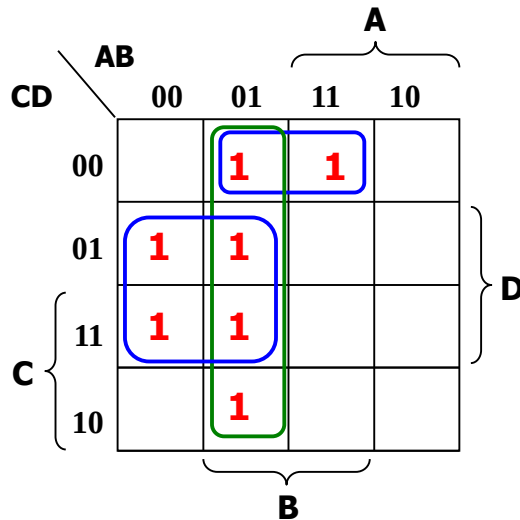
$$f(A,B,C,D) = B'.D' + B.C.D$$



# Examples

- To find simplest POS expression for example #2:  

$$f(A,B,C,D) = A.B.C + B'.C.D' + A.D + B'.C'.D'$$
- Draw the K-map of the complement of f, f'.

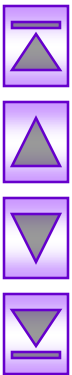


From K-map,

$$f' = A'.B + A'.D + B.C'.D'$$

Using DeMorgan's theorem,

$$\begin{aligned} f &= (A'.B + A'.D + B.C'.D')' \\ &= (A+B').(A+D').(B'+C+D) \end{aligned}$$



# Examples

- To find simplest POS expression for example #3:

$$f(A,B,C,D) = \sum m(2,8,10,15) + \sum d(0,1,3,7)$$

- Draw the K-map of the complement of f, f'.

$$f'(A,B,C,D) = \sum m(4,5,6,9,11,12,13,14) + \sum d(0,1,3,7)$$

|   |    |    |    |    |    |    |
|---|----|----|----|----|----|----|
|   |    | A  |    |    |    |    |
|   |    | AB |    | 01 | 11 | 10 |
| C | CD | 00 | 01 | 11 | 10 |    |
|   | 00 | X  | 1  | 1  |    |    |
|   | 01 | X  | 1  | 1  | 1  |    |
|   | 11 | X  | X  |    | 1  |    |
|   | 10 |    | 1  | 1  |    |    |
|   |    | B  |    |    |    |    |

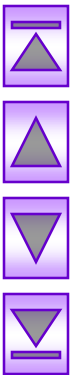
Diagram illustrating the K-map for the complement function  $f'$ . The K-map is a 4x4 grid with rows labeled CD (00, 01, 11, 10) and columns labeled AB (00, 01, 11, 10). The function  $f'$  is represented by 1s in the following cells: (00,01), (00,11), (01,01), (01,11), (11,10), (10,01), (10,11), and (10,10). The function  $f$  is represented by Xs in the following cells: (00,00), (01,00), (11,00), (11,01), and (11,11). The K-map is partitioned into four groups: a 2x2 group of 1s (blue box), a 2x2 group of 1s (yellow box), a 2x2 group of 1s (yellow box), and a 2x2 group of 1s (yellow box). The groups are labeled A, B, C, and D.

From K-map,

$$f' = B.C' + B.D' + B'.D$$

Using DeMorgan's theorem,

$$\begin{aligned} f &= (B.C' + B.D' + B'.D)' \\ &= (B'+C).(B'+D).(B+D') \end{aligned}$$



# End of segment

